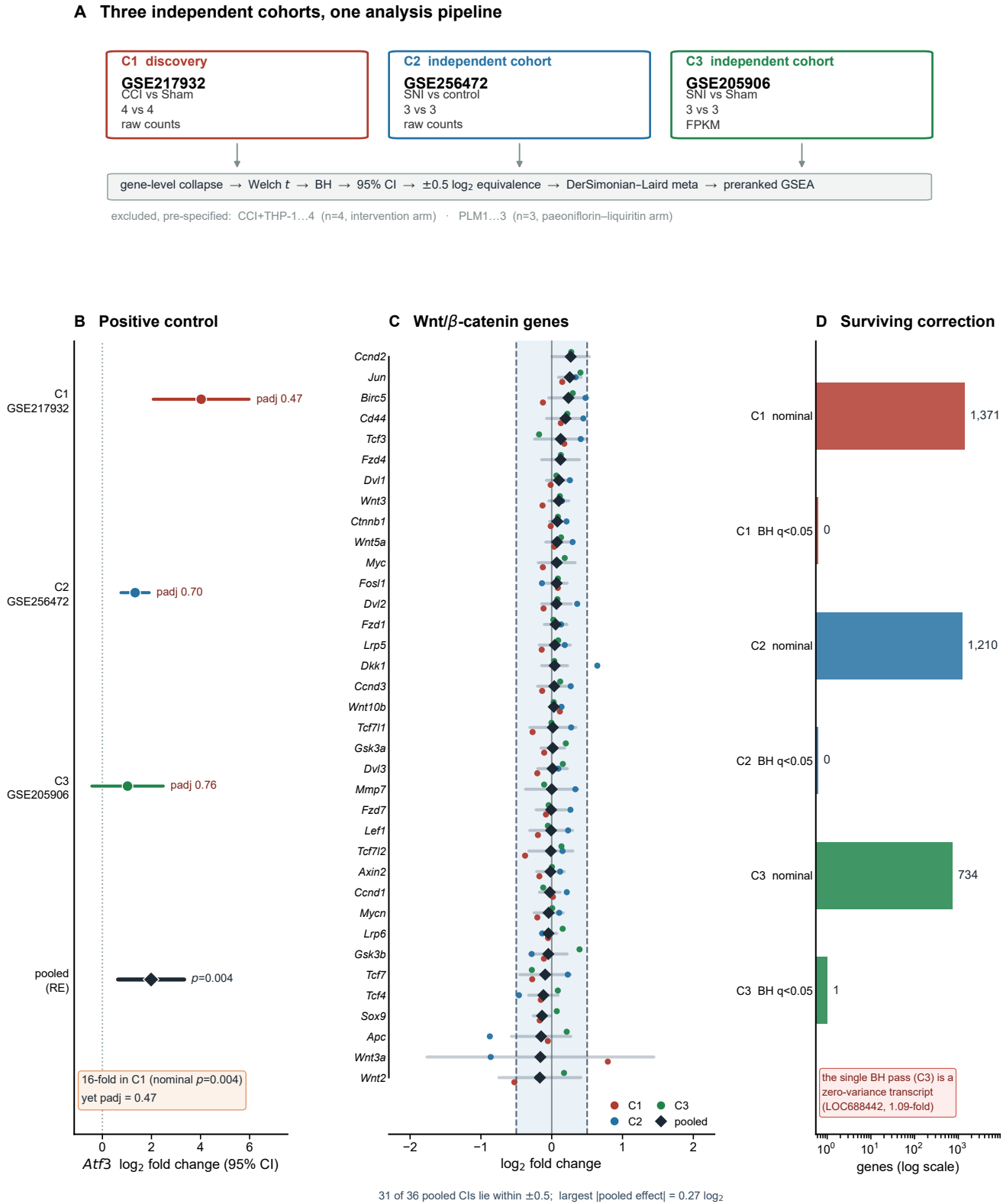

A Three independent cohorts, one analysis pipeline
C1 discovery
GSE217932
CCI vs Sham
4 vs 4
raw counts
C2 independent cohort
GSE256472
SNI vs control
3 vs 3
raw counts
C3 independent cohort
GSE205906
SNI vs Sham
3 vs 3
FPKM
gene-level collapse → Welch t → BH → 95% CI → ±0.5 log2 equivalence → DerSimonian–Laird meta → preranked GSEA
excluded, pre-specified: CCI+THP-1…4 (n=4, intervention arm) · PLM1…3 (n=3, paeoniflorin–liquiritin arm)
B Positive control
C1 GSE217932
padj 0.47
C2 GSE256472
padj 0.70
C3 GSE205906
padj 0.76
pooled (RE)
p=0.004
16-fold in C1 (nominal p=0.004) yet padj = 0.47
0
2
4
6
Atf3 log2 fold change (95% CI)
C Wnt/β-catenin genes
Ccnd2
Jun
Birc5
Cd44
Tcf3
Fzd4
Dvl1
Wnt3
Ctnnb1
Wnt5a
Myc
Fosl1
Dvl2
Fzd1
Lrp5
Dkk1
Ccnd3
Wnt10b
Tcf7l1
Gsk3a
Dvl3
Mmp7
Fzd7
Lef1
Tcf7l2
Axin2
Ccnd1
Mycn
Lrp6
Gsk3b
Tcf7
Tcf4
Sox9
Apc
Wnt3a
Wnt2
C1
C2
C3
pooled
−2
−1
0
1
2
log2 fold change
D Surviving correction
C1 nominal
1,371
C1 BH q<0.05
0
C2 nominal
1,210
C2 BH q<0.05
0
C3 nominal
734
C3 BH q<0.05
1
the single BH pass (C3) is a zero-variance transcript (LOC688442, 1.09-fold)
10^0
10^1
10^2
10^3
genes (log scale)
31 of 36 pooled CIs lie within ±0.5; largest |pooled effect| = 0.27 log2